
```

% Ezinne Akabueze, CMOR 220, Spring 2026, Project 1: Grow a Fern
% Project1_Fern.m
% This coding program is tasked to create 2 different fern plots using a
repetitive iterative
% process that includes matrix multiplication
% Last modified: 2/15/26

clc;
clear;

disp('Project 1')

% Defining all necessary functions for Project 1
% Function for Part 1 that repeats plotting an updating point z,
% eventually creating the simple fern, by analyzing the random number at 2
% different conditions
function PartOne;
    z = [1; 0]; % Starting point
    figure; hold on; grid on; grid minor; % Setting grid conditions for the
figure
    for i = 1:4000 % Use 4000 iterations to change z and plot various points
        r = rand(); % Random number to determine which condition were
evaluating from
        if r < 0.3994 % The iteration if the random number generated is less
than 0.3994
            z = [0.4 -0.3733; 0.06 0.60]*z + [0.3533; 0];
        else %The iteration if the random number generated is equal or
greater than 0.3994
            z = [-0.8 -0.1867; 0.1371 0.8]*z + [1.1; 0.1];
        end
        plot(z(1), z(2), '.', 'MarkerSize', 2.5)
    end
end

% Function for Part 2 that repeats plotting an updating point z,
% eventually creating the advanced fern, by analyzing the random number at 4
% different conditions
function PartTwo;
    z = [1; 0]; % Starting point
    figure; hold on; grid on; grid minor; % Setting grid conditions for the
figures
    for i = 1:4000; % Use 4000 iterations to change z and plot various points
        r = rand(); % Random number to determine which condition were
evaluating from
        if r < 0.01; % The iteration if the random number generated is less
than 0.01
            z = [0 0; 0 0.16]*z;
        elseif r < 0.76; % The iteration if the random number generated is
less than 0.76
            z = [0.85 0.04; -0.04 0.85]*z + [0; 1.6];
        elseif r < 0.88; % The iteration if the random number generated is
less than 0.88

```

```
        z = [0.2 -0.26; 0.23 0.22]*z + [0;1.6];
    else; % The iteration if the random number generated is greter than
or equal to 0.88, aka if probability is 0.12
        z = [-0.15 0.28; 0.26 0.24]*z + [0; 0.44];
    end
    plot(z(1), z(2), '.', 'MarkerSize', 2.5)
end
end

% Part 1
disp('Part 1, Simple Fern')
% run the function PartOne with no inputs:
PartOne

% Part 2
disp('Part 2, Advanced Fern')
% run the function PartTwo with no inputs:
PartTwo

Project 1
Part 1, Simple Fern
Part 2, Advanced Fern
```

Published with MATLAB® R2025b